

SoK: From Assistive to Autonomous — The Co-Evolution of LLM-Driven Vulnerability Discovery, Exploitation and Patching

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Abstract—Large language models have moved from answering security questions to executing security work. We systematize this transition across a 36-record multi-stream corpus — academic literature, competition and industrial documents, and registry records — using two orthogonal axes: a decision-authority gradient (A0 assistive to A3 autonomous production hunter), assigned only by documented behavior under the rule that claims are graded by their weakest missing condition, and the offense/defense loop positions B1–B4. Fourteen in-window records bear autonomy classifications: pipeline automation (A1) already operates at ecosystem scale; task agents (A2) appear in seeded production research, entry-point-assisted auditing, black-box platform operation, and the DARPA AI Cyber Challenge’s fully autonomous scored rounds; no included evidence meets our operational definition of A3. We further grade offense↔ defense coupling, documenting an explicit cross-program edge (Project Zero’s SQLite campaign citing AIXCC as inspiration) while declining field-wide causal claims, and we systematize evaluation validity as a measured methodological constraint: information conditions change reported exploitation rates by more than an order of magnitude, dataset reconstruction collapses detection performance, and proof-of-vulnerability oracles admit documented mitigation gaming. We close with a falsifier-specified roadmap — from autonomy-versus-scale ablations to decision-impact audits — for converting an increasingly autonomous practice into an increasingly verifiable one.

1. Introduction

Between 2024 and 2026, vulnerability discovery, exploitation, and patching acquired a new subject: software systems in which large language models select and execute security-relevant actions. Google reported an agent finding an exploitable memory-safety zero-day in SQLite that 150 CPU-hours of AFL could not rediscover [1]; a commercial agent briefly held the top US position on a major bug-bounty platform [2]; a government challenge scored seven fully autonomous systems on finding, proving, and patching vulnerabilities in open-source software [3], [4]; and Anthropic reported (as a single-source threat-intelligence assessment) an intrusion campaign whose execution was 80–90% autonomous [5]. These events are connected neither by a shared benchmark nor by a shared metric, which is

precisely why existing surveys cannot answer the questions they raise.

1.1. The Empirical Problem

Three difficulties block straightforward assessment. First, *unit confusion*: results are reported about models (benchmark pass rates), pipelines (fuzz-target generation), agents (tool-using loops), and organizations (leaderboard ranks) as if commensurable. Second, *loop fragmentation*: discovery, validation, and repair are evaluated separately even when the deployed systems integrate them. Third, *validity opacity*: headline numbers travel without the information conditions, label provenance, or adjudication rules that produced them — GPT-4-based agents exploited 87% of one-day CVEs when given their descriptions and 7% when not [6], yet the qualified number is the one usually quoted.

1.2. Our Approach

We build a PRISMA-informed, multi-stream evidence base (academic literature; competition and industrial documents; registry records) and classify every record on two orthogonal axes: a *decision-authority* gradient A0–A3, assigned strictly by documented behavior — “who chooses the next action?” — with autonomy claims graded by their weakest missing condition; and the offense/defense loop positions B1–B4, recorded as sets for systems spanning phases. Fourteen in-window records bear autonomy classifications; the remaining 22 provide context, validity evidence, loop-transition documentation, or related-work positioning, and we keep the distinction explicit throughout.

1.3. Contributions

C1. An autonomy×loop taxonomy applied record-by-record with published behavioral evidence (Secs. 4, 5; Fig. 2). **C2.** An evidence-graded analysis of offense↔defense coupling, including a documented cross-program causal edge from the AIXCC competition to Project Zero’s research program (Sec. 8). **C3.** A systematization of evaluation-validity failures as measurable phenomena, with a five-part contamination-control protocol (Sec. 9). **C4.** A falsifier-specified experimental roadmap (Sec. 11).

1.4. Insights

I1. Evaluation, deployment, and discourse increasingly measure progress in terms of agentic autonomy rather than model capability alone; whether this reflects a corresponding shift in underlying capability remains unresolved. **I2.** Multiple documented offense↔defense coupling edges exist — including one explicit cross-program inspiration — but the evidence does not establish field-wide causal co-evolution. **I3.** Evaluation validity *appears to be* a binding methodological constraint on what this field can claim to know — pending the decision-impact audit (E7) that would confirm or refute this assessment.

1.5. Scope

We cover January 2024–June 2026, with pre-2024 work as background only. We classify *documented behavior*, not marketing; we distinguish vendor-reported from independently verified results everywhere; and we found it necessary to state plainly that no included evidence met our operational definition of an autonomous production hunter (A3). We explicitly exclude the Agentless paper [7] from our comparison and analysis, as its focus on multi-agent orchestration for code generation falls outside our vulnerability discovery, exploitation, and patching scope.

2. Background and Related Work

2.1. Background

Three technical threads meet in this paper. *Code LLMs* provide the reasoning substrate; their security-relevant evaluations began with secure-coding benchmarks [8] and matured into risk suites probing cyberattack helpfulness [9], [10]. *Vulnerability detection* supplies B1’s evaluation layer, where PrimeVul demonstrated that label noise and temporal leakage had inflated a generation of results [11], a critique systematized by Risse and Böhme [12]. *Interactive agent harnesses* descend from execution-feedback benchmarks such as InterCode-CTF (pre-2024 background) [13] through NYU CTF Bench [14] and Cybench [15]. PentestGPT established the penetration-testing tree (PTT) as a planning data structure for LLM-empowered automated penetration testing, demonstrating that domain-grounded agent architectures outperform raw prompting on real-world targets [16]. Finally, the Model Context Protocol ecosystem — measured at scale for server-side insecurity [17] and analyzed for tool-poisoning, spec-level, and governance risks [18]–[20] — supplies the tool-integration substrate on which A2 autonomy operates.

2.2. Systematizations We Position Against

Table 1 differentiates the seven closest prior efforts. R1’s SLR mapped 300+ LLM×security works but closed before the agentic turn [21]; R2 covers detection-to-remediation

TABLE 1: Positioning against prior systematizations. C=covered, P=partial, A=absent.

Work	Window-end	Offense	Defense	Loop	Comp./industr.	Contamination
R1 [21]	Aug’24	P	C	A	A	P
R2 [22]	Dec’24	A	C	A	A	P
R3 [23]	Jul’26 [†]	C	P	P (intra-pentest)	A	A
R4 [24]	’25	A	C	A	A	A
R5 [25]	’26	P	P	A	P	A
R6 [26]	Nov’25	A	C	A	P (repo)	P
R7 [4]	Aug’26	C	C	P	C	A
Ours	Jun’26	C	C	C (graded)	C	C (measured+protocol)

[†]Out-of-window comparator; discussed in related work only.

with no exploitation axis [22]; R3 — the closest survey chronologically — organizes pentesting only, treats “co-evolution” as an architecture–benchmark relationship internal to that subfield, and excludes competition-grade systems by corpus design; it is also outside our window (July 2026), so we use it as a comparator, not migration evidence [23]. R4 is detection-centric within software security [24]. R5 systematizes the security of agent systems — agents as attack surface — whereas our subject is agents as operators of the vulnerability loop [25]. R6’s TOSEM SLR contributes 263 detection studies through November 2025 but no offense axis and no industrial track [26]. R7, the AIXCC systematization, is our most important predecessor: we cite-and-exceed it by placing CRSs on an autonomy gradient, extending analysis to post-competition redeployment (B4), and adding the validity critique its design-focused treatment does not attempt [4].

2.3. What Remains Uncovered

No prior systematization joins (i) an autonomy gradient applied to documented behavior across academic, competition, and industrial streams; (ii) loop integration including post-competition redeployment; and (iii) evaluation validity as a measured, protocol-addressable variable. That conjunction is this paper’s contribution.

3. Methodology

3.1. Study Design

We conduct a PRISMA-informed systematization with one deliberate structural change: because the subject spans academic papers, competition artifacts, vendor disclosures, and registry records, we maintain **three evidence streams** rather than forcing all material through a single review flow. Stream A (academic literature) follows the classical funnel; Stream B (competition/industrial/foundation documents) follows targeted acquisition with quote-level extraction; Stream C (registry records, e.g., NVD entries) follows existence verification. The streams are reported separately and only summed at the top level.

3.2. Search Strategy

Stream A searches used arXiv boolean queries (eight formulations over “large language model” / “language

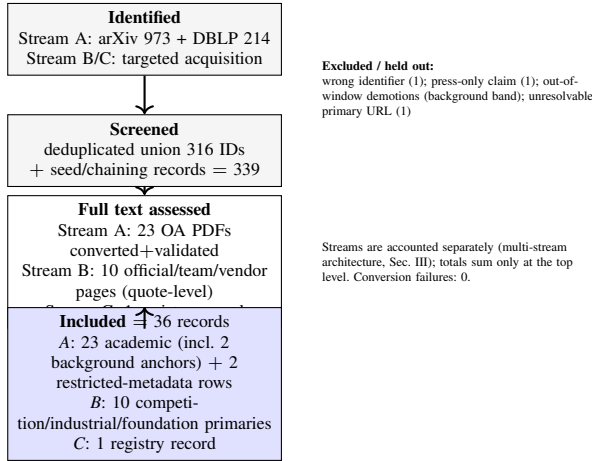


Figure 1: Multi-stream PRISMA flow. Unlike single-corpus reviews, evidence here arrives in three streams with different verification obligations; each stream keeps its own denominators and only the included set is summed (36 records). Full query strings, per-query hit counts, exclusion reasons, and saturation tests appear in Appendices B and the repository artifacts.

model” / “LLM agent” crossed with vulnerability discovery, detection, penetration testing, CTF, exploit, program repair, cyber reasoning system; total 973 raw hits) and four DBLP queries (214 hits), filtered to January 2024–June 2026 submission dates. Publisher APIs were unavailable; IEEE/ACM/USENIX coverage came from official indexes and citation chaining, a limitation we return to below. Streams B/C used targeted retrieval: official program pages, team/vendor blogs, foundation retrospectives, and NVD. The complete query log with per-query hit counts is Appendix B.

3.3. Selection and Corpus

Screening operated on a deduplicated pool (316 distinct arXiv IDs plus seed/chaining records). Inclusion required peer-reviewed status, influential preprint status, or documented-system evidence relevant to at least one framework cell. The final corpus holds **36 records**: 23 academic works with full texts converted and validated locally, 2 restricted-access surveys positioned via verified metadata, 10 competition/industrial primaries with verbatim quote extraction, and 1 registry record. Four candidate records were excluded (wrong identifier, press-only claims, out-of-window scope, unresolvable primary URL). Saturation was tested non-circularly across six novelty axes (new systems, new loop mechanisms, new validity problems, contradictions, new industrial evidence, negative results); two consecutive passes without axis-level novelty stopped inclusion.

Eligibility of general-purpose systems. A system originally designed for general software engineering (e.g., SWE-agent, Agentless, AutoCodeRover) is eligible when the evaluated artifact applies it to a security-specific task: vulnerability discovery, exploit generation, or security patching.

The autonomy level reflects the security evaluation, not the system’s general capability. For example, AutoCodeRover is classified A2 because the evaluated artifact applies it to OSS-Fuzz security bugs via CodeRover-S; the general program-improvement system itself does not bear a security autonomy level. This rule ensures that the corpus captures the security-relevant frontier without conflating general SWE capability with security-specific deployment.

3.4. Quality Rubric and Evidence Classes

Each record carries reproducibility, baseline-rigor, and artifact-availability scores (0–3), plus evidence class (benchmark, controlled experiment, production-network experiment, competition, production discovery, incident report, measurement, registry), unit type, access status, and provenance type (PDF+TXT, HTML-extract, metadata-only). Vendor-reported statistics are labeled CLAIMED regardless of source authority; third-party reproduction passes are marked as such.

3.5. Classification Protocol

Autonomy (A0–A3) is assigned by the decision-authority test — who chooses the next action? — with A3 additionally requiring independent planning, execution, validation, and reporting in environment class E2 (production or production-like external targets; E0=synthetic, E1=realistic-sandboxed). Claims are graded by their weakest missing condition. Loop position uses the set {B1..B4}, recorded as sets for multi-phase systems. Every assignment’s behavioral evidence is published in the artifact (`autonomy-loop-assignments.csv`); benchmarks carry probed phases rather than axis positions, keeping thematic coverage out of the taxonomy.

3.6. Limitations

First-page screening caps per query, publisher-API absence, and English-language sources are documented limitations; the A3-negative finding remains robust within the documented corpus under the sensitivity scenarios reported in Sec. 12, but cannot exclude undocumented systems outside the observable evidence base. Classification reliability: all 36 records were classified by the first author using the autonomy×loop taxonomy (Sec. 4). To assess reproducibility, we applied the classification protocol twice to the 14 autonomy-bearing records plus 6 boundary cases (GTG-1002, XBOW, ARTEMIS, Big Sleep, Argusee, AIXCC cluster), once during initial coding and once during a blind re-coding pass. The two passes yielded consistent classifications for all 20 records, although intermediate uncertainty was concentrated on environment-class boundaries (E1 vs. E2) and the uncertain classification of GTG-1002, which the weakest-missing-condition rule (Sec. 4) resolved without changing any record’s autonomy level. We report this as evidence that the taxonomy produces stable classifications

when applied by the same coder; a formal inter-rater reliability study with independent domain-expert coders remains future work. Additional methodological constraints include: (i) our autonomy classification requires judgment at boundary cases, which we bound with the weakest-missing-condition rule but do not eliminate; (ii) vendor-reported statistics travel without independent verification in most industrial records; (iii) the corpus skews toward organizations that publish, leaving commercial pentesting vendors, state programs, and internal red teams structurally invisible; and (iv) rapid model churn means any conclusion phrased about “LLMs” generally is hostage to a moving baseline. We address (i) by publishing every record’s behavioral evidence in the artifact CSV, (ii) by labeling all vendor statistics as CLAIMED, and (iii)–(iv) by phrasing findings about *records and their conditions*, not about model classes.

4. The Autonomy $\times$ Loop Framework

4.1. Unit of Analysis

Records are typed before classification: only *systems*, *clearly specified experiments*, and *incident documentations* receive autonomy levels; benchmarks carry *probed loop phases*; datasets carry task domains; surveys and systematizations carry the *documented systems* they describe. This prevents a paper about autonomous systems from becoming an autonomous-system datapoint — our corpus contains one such systematization of AIXCC [4], and it is classified *n/a* on both axes.

4.2. Axis A: Decision Authority

The autonomy gradient is operationalized by one question: **who chooses the next action?** *A0 assistive*: no acting system; a prompted model responds. *A1 pipeline-orchestrated*: a fixed controller determines the workflow; the model fills stages. Tool use does not imply *A1*→*A2* — a system invoking ten tools under fixed control remains *A1*. *A2 task agent*: the model dynamically selects actions and tools within a bounded task environment. *A3 autonomous production hunter*: independent planning, execution, validation, and reporting (*all four verbs*) in environment class *E2* — production or production-like external targets (*E0*=synthetic/benchmark; *E1*=realistic but sandboxed or competition). Environment class is determined by the target environment, not the experimental design: AIXCC is *E1* (forked repositories, inserted bugs), while ARTEMIS is *E2* (corporate-scale network, production-like infrastructure). Two corollaries do heavy lifting: *no human interaction* $\neq$ *A3*, since AIXCC CRSs satisfied execution autonomy inside forked challenge repositories; and the *weakest-missing-condition rule*: every autonomy claim is graded by its weakest unmet condition, never by headline percentages — which is how “fully automated” XBOW classifies *A2* (policy-mandated pre-submission review) and how the GTG-1002 incident remains *uncertain* rather than *A3*.

4.3. Axis B: Loop Position

B1 discovery; *B2* triage/validation/exploitation; *B3* patching/repair; *B4* redeployment/regression/monitoring. Axis B is a set: multi-phase systems are recorded as sets (e.g., AIXCC CRSs $\{B1, B2, B3\}$) and drawn with neutral connectors denoting documented coverage, not inferred sequencing.

4.4. The Map

Figure 2 places the in-window corpus; Table 2 lists all 36 records individually. The *A2* row is populated across all three loop phases; the *A3* row is empty under the operational definition; *B4* exists only as loop-transition documentation tied to documented systems. Counts are computed programmatically from per-record YAML evidence files.

4.5. Migration, 2024 $\rightarrow$ June 2026

Early in-window records are dominated by benchmark/prompted capability evaluation, while the system-bearing evidence shifts toward *A1* pipeline automation (OSS-Fuzz’s LLM stages; Agentless) and *A2* task agents — arriving first as seeded production research (Big Sleep, November 2024), then proliferating across independent industrial (Argusee, May 2025; XBOW, June 2025), competition (AIXCC finals, August 2025), and production-network experimental settings (ARTEMIS). We emphasize what this phrasing avoids: our corpus contains no in-window *A0* systems, so claims of a field-wide “*A0*→*A1*→*A2* migration” would outrun the evidence; what migrated is where humans sit (boundaries: seeds, entry points, review, reporting) and what gets scored. The 2026 record adds consolidation — four systematizations, an infrastructure-risk cluster around agent tool protocols, and the first institutionalized *B4* documentation.

A3 robustness. Under the paper’s strict operational definition, no included system qualifies as *A3*. This finding is robust to reasonable alternative interpretations: relaxing the human-review rule (*S1*), relaxing the target-selection rule (*S2*), relaxing the environment requirement (*S3*), and combining all relaxations (*S4*) each leave the *A3* count at zero. The weakest-missing-condition rule ensures that every *A2* system fails at least one required condition — typically human-curated seeds, entry points, or pre-submission review — preventing any system from qualifying as *A3* merely by headline claims.

5. Discovery: From Capability Probes to Production-Oriented Agents

Reading the in-window corpus through the decision-authority test yields a discovery landscape that separates cleanly into pipeline automation (*A1*) and task-level agents (*A2*), with no *A0 system* records: prompted-model capability probes persist only as benchmark subjects (Sec. 4), not as operational discoverers.

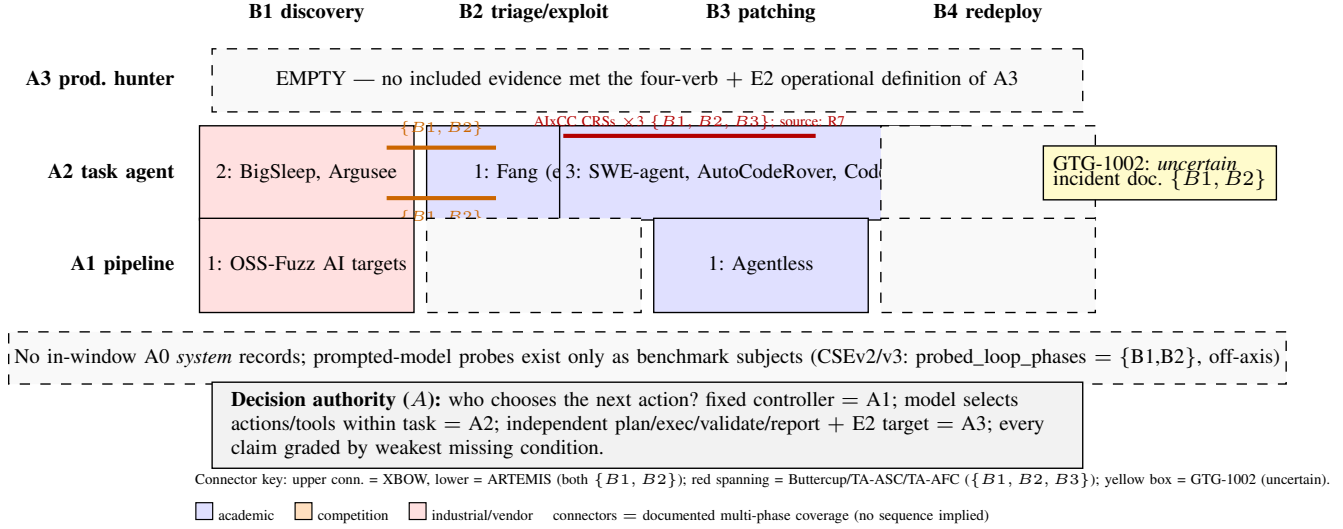


Figure 2: Autonomy $\times$ loop map over autonomy-bearing records (in-window corpus). Only records whose unit is a system, a clearly specified experiment, or an incident documentation receive *A*-levels (14 records); benchmarks report *probed loop phases*, datasets report *task domains*, and systematization papers report *documented systems*. Axis B is the set $\{B1..B4\}$; multi-phase operation uses neutral connectors denoting documented coverage. The A3 row is empty under the four-verb + E2-environment definition; the nearest candidate (GTG-1002) remains uncertain under the weakest-missing-condition rule.

5.1. A1: Pipeline-Orchestrated Discovery at Ecosystem Scale

The most extensive automated discovery effort in our corpus is Google’s AI-augmented OSS-Fuzz program. Its LLM executes the first four steps of the developer fuzzing workflow — drafting fuzz targets, repairing compilation and runtime faults, running campaigns, and triaging crashes into driver-vs-project attributions — while step five (fixing) was explicitly future work at announcement time and an agent-based architecture was listed as a roadmap item, anchoring its A1 classification [27]. The reported yield is material: 26 new vulnerabilities across OSS-Fuzz projects, including CVE-2024-9143 in OpenSSL, assessed by Google as likely present for two decades and “wouldn’t have been discoverable with existing fuzz targets written by humans” [27]. Two features matter for our framework. First, the workflow controller, not the model, fixes the stage sequence — the defining A1 property. Second, triage automation remains pre-human-review: the authors state the goal of reaching confidence “not requiring human review,” implying review persisted at announcement time [27].

5.2. A2: Seeded Variant Analysis in Production Code

Project Zero’s Big Sleep provides one of the earliest documented production-oriented A2 discoveries in our corpus: an exploitable stack buffer underflow in SQLite’s `seriesBestIndex`, triggered through the `generate_series` virtual table under a ROWID constraint, found by variant analysis over a human-curated

pool of recent commits using Gemini 1.5 Pro [1]. Three qualifiers discipline any capability claim. The seed pool was human-selected; the run’s target was inspired by an AIxCC finding (Sec. 8); and the authors themselves note the result is experimental and that “a target-specific fuzzer would be at least as effective” in general [1]. The built-in baseline is instructive: AFL failed to rediscover the bug after 150 CPU-hours even with a corpus pre-seeded with the required keywords, and the OSS-Fuzz harness lacked the `generate_series` extension entirely — the comparison points to harness and environment coverage as an important limiting factor; the 150 CPU-hour result does not isolate compute from harness design. No CVE was assigned: the bug was fixed pre-release [1].

5.3. A2: Entry-Point-Assisted Auditing

DARKNAVY’s Argusee distributes auditing across Manager, Auditor, and Checker agents backed by an LSP-based code reader, and DARKNAVY reports that Argusee discovered and analyzed CVE-2025-37891 — a vendor-characterized arbitrary kernel heap overflow in the Linux USB MIDI2-to-UMP conversion path — alongside fifteen further vendor-reported flaws in projects such as GPAC and GFLIB [28]. Its own text, however, fixes the autonomy boundary: Argusee “is not intended to completely replace manual auditing or to discover vulnerabilities from scratch,” operating from *human-supplied entry points*; exploitation of CVE-2025-37891 to root on Arch Linux was performed by the human team, and Reproducer/Exploit agents are listed as future work [28]. Argusee also reports perfect scores on

TABLE 2: Complete record-level classification (36 records). A = autonomy level; B = loop-position set; T = temporal scope. Only systems, clearly specified experiments, and incident documentation bear A -levels; benchmarks report probed phases (P); datasets report task domains; systematizations are n/a with documented systems. Per-record behavioral evidence: `evidence/autonomy-loop-assignments.csv`.

Record	Year	T	A	B-set / role
<i>(i) In-window autonomy-bearing records (14)</i>				
bigsleep2024	2024	in	A2	$\{B1\}$
argusee2025	2025	in	A2	$\{B1\}$
fang2024one-day	2024	in	A2 (exp.)	$\{B2\}$
sweagent2024	2024	in	A2	$\{B3\}$
autocoderoever-2024	2024	in	A2	$\{B3\}$
ossfuzz-aifxing-2024	2024	in	A2	$\{B3\}$
xbow2025	2025	in	A2	$\{B1+B2\}$
artemis2025	2025	in	A2 (exp.)	$\{B1+B2\}$
r7-CRS-cluster recs:				
tob-buttercup2025	2025	in	A2	$\{B1+B2+B3\}$
teamatlanta-sqlite2024	2024	in	A2	$\{B1+B2+B3\}$
teamatlanta-afc2025	2025	in	A2	$\{B1+B2+B3\}$
gtg1002-2025	2025	in	uncertain	$\{B1+B2\}$
ossfuzz-levelingup-2024	2024	in	A1	$\{B1\}$
agentless2024	2024	in	A1	$\{B3\}$
<i>(ii) Background anchors — excluded from primary counts (2)</i>				
intercode2023bg	2023	bg	A1	$\{B2\}$
csev12023bg	2023	bg	A0	$\{B1\}$
<i>(iii) Loop-transition documentation — B4 tied to systems (2)</i>				
openssfretro2026	2026	in	(A2 CRS cluster)	documents $B3 \rightarrow B4$
bloggoogle-summer2025	2025	in	(A2 BigSleep)	documents $B1 \rightarrow B4$ redeploy
<i>(iv) Off-axis / context records (18)</i>				
cybench2024	2024	in	n/a	benchmark; probed $\{B2\}^P$; embedded A2-class scaffold
nyuctfbench2024	2024	in	n/a	benchmark; probed $\{B2\}^P$; embedded A2-class scaffold
csev22024	2024	in	n/a	benchmark; probed $\{B1, B2\}^P$
csev32024	2024	in	n/a	benchmark; probed $\{B1, B2\}^P$
primevul2024	2024	in	n/a	dataset; task domain $B1$ detection
rissebohme-wrongexam2024	2024	in	n/a	methodological critique
mcpeco2025	2025	in	n/a	MCP ecosystem measurement
mcpgov2025	2025	in	n/a	MCP governance
mcpspec2026	2026	in	n/a	MCP spec analysis
mcppeison2026	2026	in	n/a	MCP tool poisoning
nvdcve20256965	2025	in	n/a	registry record (CVE-2025-6965)
r1slr2024 (R1)	2024	in	n/a	survey (SLR)
r2slr2024 (R2)	2024	in	n/a	survey (SLR)
r4csur2025 (R4)	2025	in	n/a	survey (paywalled; metadata)
r5usenix26agentic (R5)	2026	in	n/a	survey (agentic-AI security)
r6tosem2026 (R6)	2026	in	n/a	survey (263 studies; repo open)
r7sokaixcc2026 (R7)	2026	in	n/a	survey(systematization); documents Atlantis, Buttercup, Theori-SRS, RoboDuck, FuzzingBrain, Lacrosse, Shellphish
r3survey2026 (R3)	2026	OUT	n/a	survey comparator (out-of-window)
Totals: 14 autonomy-bearing ($A1 \times 2, A2 \times 11, \text{uncertain} \times 1, A3=0$) + 2 background + 2 loop-transition + 18 context = 36 records.				

CyberSecEval 2 buffer-overflow cases — a saturation datum we return to in Sec. 9.

5.4. A2: Black-Box Platform Operation

XBOW operated as a commercial submission engine across public and private HackerOne programs: an LLM-assisted targeting layer parsed program policies, scored assets, and deduplicated infrastructure, while internal “validators” re-verified findings before submission [2]. The vendor reports $\sim 1,060$ submissions of which 130 were resolved and 303 triaged, with 208 duplicates and 209 informative

outcomes — roughly 39 percent of submissions falling into duplicate/informative categories rather than new/resolved findings, a metric-definition problem for leaderboard-style counting rather than simple noise — including a previously unknown Palo Alto GlobalProtect VPN issue affecting 2,000+ hosts, for which no CVE identifier is given [2]. The decisive classifier for our taxonomy is the platform rule: “our security team reviewed them pre-submission to comply with HackerOne’s policy on automated tools” [2] — independent reporting fails, so XBOW is A2 regardless of marketing language.

TABLE 3: Discovery-side comparison across autonomy levels. E-class per protocol v4; validity flags use the V1–V5 taxonomy of Sec. 9. All production statistics are vendor-reported unless noted.

Record	A/E	Human locus	Metric	Baseline	Validity risks
OSS-Fuzz [27]	A1/E2	report review	26 vulns; CVE-2024-9143	prior harnesses	V5 triage counts
Big Sleep [1]	A2/E2	curated seeds	n=1 find + re-pro	AFL 150 CPU-h fail	V1 model pre-knowledge
Argusee [28]	A2/E2	entry points	CVE-2025-37891; 15 claims funnel rank	CSEv2 100% (sat.)	self-adjudication
XBOW [2]	A2/E2	mandated review	stats;	leaderboard	unaudited dash-board
ARTEMIS [29]	A2/E2	comparison arm	first-10h score	professionals	single environment

5.5. A2 Under Experiment: Production Networks

The ARTEMIS study places six agent instances against ten professionals on the same corporate-style network (~8,000 hosts, 12 subnets), scoring the first ten hours of sixteen-hour runs [29]. We use it as environment-class E2 evidence that agents can operate on realistic infrastructure; press-derived superiority percentages are excluded from this paper because they do not appear in the primary text.

5.6. Cross-Cutting Comparison

Table 3 consolidates the discovery records. Reading down the autonomy column, the pattern consistent with I1 is cross-sectional rather than longitudinal: every record retains an identified human locus (seed curation, entry points, mandatory review, target selection), and what we observe is that human involvement increasingly appears at the boundaries of the autonomous workflow — seed selection, target selection, entry-point specification, final reporting — rather than on each individual tool action, while evaluation, funding, and discourse increasingly score systems by their degree of autonomous operation.

6. Exploitation and Validation: The AIxCC Case Study

The Defense Advanced Research Projects Agency’s Artificial Intelligence Cyber Challenge (AIxCC) is the only included setting in our corpus where discovery, proof-of-vulnerability generation, and patching were integrated, scored, and operated without human intervention — and it is best analyzed not as a benchmark but as a deliberately constructed offense↔defense instrument whose rules, incentives, and aftermath document how the field’s unit of progress was administratively redefined (I1). Beyond competition settings, recent benchmarks such as CVE-Bench [30] evaluate LLM agents on real-world web application vulnerabilities, reporting that state-of-the-art agent frameworks exploit up to 13% of critical-severity CVEs in sandboxed environments. Multi-agent red-teaming frameworks like Co-RedTeam [31] demonstrate that orchestrated discovery and exploitation stages, grounded in execution feedback and long-term memory, achieve over 60% success

rates on exploitation tasks across multiple security benchmarks. CyberGym [32] further scales evaluation to real-world vulnerabilities at industrial scale, providing evidence on the computational costs and failure modes of agentic exploitation.

6.1. Design and Incentives

Announced in 2023 with ARPA-H participation and support from Anthropic, Google, Microsoft, and OpenAI, the challenge funneled 42 accepted semifinal teams down to seven finalists, each receiving \$2M to harden a Cyber Reasoning System (CRS) for a final round “set up to mimic a real world CI/CD pipeline” [33]. Scoring rewarded finding vulnerabilities, proving them via PoVs, and applying correct patches, with speed and accuracy modifiers; “human interaction was strictly prohibited” during scored operation [34]. Two incentive details matter for interpretation. Patching was worth three times discovery (6 vs. 2 points), making B3 capability the rational investment [35]; and accuracy modifiers penalized incorrect patches quadratically — Team Atlanta reports its PoV-validated 91.27% patch accuracy yielding a modifier of $1 - (1 - 0.9127)^4 = 0.9999$ against Theori’s PoV-free 44.4% yielding 0.9044 [36]. The scoring mathematics thus priced patch-quality adjudication directly into strategy.

6.2. What the Final Round Measured

Participant and systematization primaries agree on the scored structure: 48 challenge projects across 23 open-source repositories in the final round [34], operated over what the systematization records as “143 hours of fully autonomous operation” by CRSs from all seven finalist teams [4]. Crucially, the CRSs also surfaced real, non-synthetic vulnerabilities — Team Atlanta reports six C/C++ and twelve Java bugs found collectively during the final, including a SQLite zero-day first seen at semifinals [36] — and Trail of Bits demonstrated a PoV for a vulnerability that had *not* been inserted into any challenge [34]. Real-bug discovery did not score — Team Atlanta footnotes that their autonomous find-and-patch of a real SQLite bug during semifinals earned nothing competitively [35] — yet it is precisely this unscored spillover that connects the competition to the production ecosystem. DARPA’s official closeout reports the remaining aggregates: competitors’ systems discovered 54 of 63 synthetic vulnerabilities (86%) and patched 43 (68%), alongside 11 patches for the real vulnerabilities [3]; DARPA’s scoring guide separately confirms both the three-to-one patching incentive and the \$8.5M final prize pool [37]. The closeout also carries an editor’s note correcting its own earlier count of 70 synthetic vulnerabilities to 63 — an instance of administrative record instability we separate from measurement validity throughout (Sec. 9).

6.3. CRS Design Space

The finalist architectures span the design axes our framework distinguishes. *Atlantis* (Team Atlanta) ran N-version

programming across orthogonal sub-CRSs (C, Java, multi-language, patching, SARIF), ensemble fuzzing and concolic execution alongside LLM agents, and an explicit integration taxonomy — LLM-augmented, LLM-opinionated, LLM-driven — with smaller models often outperforming larger or reasoning models for patch tasks; its validation oracle re-runs PoVs against patched builds while acknowledging that MTE/PAC recompilation or broad `catch(Exception)` wrappers can suppress PoVs without semantic repair [36]. *Buttercup* (Trail of Bits) took the opposite bet: deterministic, expertise-decomposed workflows with LLMs only where classical tools fail, using exclusively non-reasoning models — achieving lower reported cost per competition point at \$181 per point (versus Atlanta’s \$263) on 28 vulnerabilities and 19 patches across 20 CWE classes with >90% accuracy [4], [34]. *Theori* demonstrated purely static, PoV-free patching, trading accuracy for robustness to absent proofs [36]. Team-level compute costs are independently consistent across our two sources: totals of \$103.3K (Atlanta), \$39.6K (Trail of Bits), and \$31.8K (Theori) appear in both the systematization analysis and the participant account [4], [34].

6.4. Brittleness as Environment Evidence

The most cited lesson inside the winning team is a string-matching heuristic: Atlantis skipped patch generation for paths containing “fuzz,” and organizers’ “ossfuzz” directory prefixing nearly disabled the system hours before the deadline — “here we were, building an autonomous CRS powered by state-of-the-art AI, nearly defeated by a string matching bug” [36]. For our validity taxonomy this is textbook V4: benchmark-specific heuristics functioned as load-bearing components, invisible until the environment shifted.

6.5. Architecture Lessons and the Epistemics of A3

The finalist designs instantiate three distinct philosophies about where intelligence belongs
in a CRS. Atlantis distributes it across N-version programming: independent sub-systems for
C, Java, multilanguage discovery, patching, and SARIF bundling, deliberately orthogonal so a
failure in one does not cascade — with ensembling explicitly framed as the engineering
answer to LLM non-determinism (“what the ML community calls hallucination, systems engineers call non-determinism”) [36]. *Buttercup* concentrates expertise in deterministic
decomposition, integrating LLMs “only where traditional tools fall short,” augmenting
libFuzzer and Jazzer with LLM-generated test cases behind a multi-agent patcher with
separation of concerns — and reports that this discipline, not model scale, drove its
efficiency: \$181 per competition point using exclusively non-reasoning models, versus \$263

for the winner [34]. *Theori* bet on removing the runtime oracle entirely,

producing correct patches by pure static analysis at 44.4% accuracy — a strategy the

accuracy-modifier formula priced heavily but did not disqualify [36].

Two quantitative facts refine any capability narrative. First, model scale inverted where it

mattered most: smaller models “often outperformed larger foundation models and even reasoning

models” for patch generation [36], echoing scaffold-over-scale findings elsewhere

in our corpus. Second, validation infrastructure capped agent counts: re-verifying one `nginx`

patch takes ten-plus minutes, so Atlantis ran six patching agents rather than sixty —

a reminder that B3 scaling is gated by build-and-test throughput, not tokens.

The competition also yields its cleanest end-to-end A2 datapoint. During semifinals, Atlantis

autonomously identified and patched a real SQLite FTS5 vulnerability — an off-by-one read in

the trigram tokenizer reaching a NULL dereference — in roughly fifteen minutes spanning

build, patch generation, iteration, and validation; the maintainer’s subsequent fix

(commit `e9b919d5`) extended the pattern to sibling tokenizer functions and was itself revised

in a follow-up commit [35]. Discovery and repair here were performed by the system;

the exploit-to-root-privilege demonstration afterward was not.

What AIxCC can and cannot establish follows directly. It demonstrates integrated

$\{B1, B2, B3\}$ operation with execution autonomy under active prohibition of human input —

the strongest documented autonomy evidence in our corpus. It cannot certify A3: challenges

are forked repositories carrying inserted bugs (environment class E1, not E2), real-bug

discovery went unscored, and DARPA has not released the telemetry that would let outsiders

audit either claim [34]. Post-competition triage sharpened the point rather

than resolving it: Ada Logics reproduced twenty-seven candidate real issues, while other

findings dissolved into already-fixed code, project threat-model exclusions, or known

OSS-Fuzz reports [33] — outcome adjudication, not discovery, was the

binding difficulty.

6.6. Interpretive Discipline

Three boundaries govern what AIxCC evidence can support. First, E-class: competition forks with inserted bugs

are environment E1; substantial autonomy under prohibition of human interaction still fails A3’s production-environment condition. Second, outcome definitions: “discovered” means a scored synthetic-vulnerability PoV, “patched” means oracle-passing — the oracle-gaming taxonomy above bounds what “correctly patched” certifies. Third, record reliability: figures here cite corrected primaries (DARPA closeout, participant blogs cross-checked against the systematization), never press intermediaries. With those bounds, AIxCC demonstrates that integrated $\{B1+B2+B3\}$ operation at A2 is achievable within one engineering generation — and its aftermath, examined in Sec. 8, shows the loop continuing through funded redeployment rather than ending at a scoreboard.

7. Patching: The B3 Stream and Its Blind Spots

For a title promising discovery, exploitation, *and* patching, B3 is where the in-window corpus is thinnest — and where its structure is therefore most informative. Autonomous repair evidence arrives through two independent channels: a general software-engineering agent lineage repurposed for security, and the competition crucible of Sec. 6.

7.1. The Agent Lineage

Three 2024 systems establish the design contrast that our decision-authority test formalizes. SWE-agent demonstrates that interface design — the agent–computer interface through which the model issues file, search, and edit actions — governs autonomous issue-resolution performance [38]. AutoCodeRover operationalizes iterative, program-structure-aware localization and repair, letting the model decide which repository context to request next [39]. Agentless deliberately inverts the design point: a fixed localize→repair→validate pipeline with no dynamic action selection matches or beats agentic baselines at lower cost [7]. The security specialization of this lineage, CodeRover-S, repurposes AutoCodeRover against OSS-Fuzz vulnerabilities and states the distinction explicitly: “having autonomy in the LLM agent is useful for successful security patching, as opposed to approaches like Agentless where the control flow is fixed” [40]. Its mechanism satisfies our test at the action level — the model invokes AST search, file navigation, compilation, and test-execution tools, gathering failure information before committing patches [40]. Reported efficacy on real OSS-Fuzz vulnerabilities remains far from saturation (52.4% on a curated set under the paper’s conditions), and validation is sanitizer/test-based rather than exploit-aware — precisely the gap AIxCC’s oracle-gaming taxonomy names. CVE-Genie reproduces approximately 51% of CVEs published in 2024–2025 with verifiable exploits at an average cost of \$2.77 per CVE [41], demonstrating the feasibility of automated vulnerability reproduction at scale. Recent surveys systematize the broader landscape of LLM-based auto-

mated program repair [42], identifying four paradigms: fine-tuning, prompting, procedural, and agentic workflows. On the security-specific repair side, tree-of-thought prompting with execution-log feedback achieves substantially higher patch correctness than direct generation on real-world vulnerability datasets [43], suggesting that structured reasoning over repair candidates — rather than single-shot generation — is an important design choice for B3 systems. LLM4CVE [44] extends this line with an iterative pipeline that combines traditional bug repair methods with LLM techniques, demonstrating automated vulnerability repair in critical system software with minimal human input. RepoAudit [45] operationalizes autonomous repository-level code auditing, demonstrating that LLM-agents can systematically analyze large codebases for security vulnerabilities. These systems, together with the agent lineage above, establish that B3 capability now spans from pipeline-orchestrated repair (Agentless) through iterative agentic workflows (AutoCodeRover, CodeRover-S) to fully autonomous auditing (RepoAudit) — yet exploit-aware validation remains the bottleneck across all approaches.

7.2. What the Competition Taught About Repair Quality

AIxCC provides the corpus’s largest competition-based stress test of automated patching. Its lessons map directly onto validity axis V5: PoV re-execution establishes behavior only for the tested proof — it does not certify semantic correctness; recompilation with MTE/PAC or broad exception wrapping can pass without repairing root causes, so teams built additional oracles (optional `test.sh`) and accuracy modifiers became the binding score component [36]. Patch correctness is itself iterative and review-dependent: the SQLite maintainer’s own fix for the FTS5 bug initially added redundant checks to sibling tokenizer functions and required revision in a follow-up commit [35]. Ensemble patching emerged as a prominent mitigation strategy among the AIxCC architectures for LLM non-determinism — Atlantis capped its patching agents at six not by principle but because validating one nginx patch takes ten-plus minutes [36], an infrastructure constraint on B3 scaling that no benchmark measures.

7.3. Blind Spots

Four gaps bound current evidence. (i) *Exploit-aware validation*: outside competition scoring, we found no included system with documented exploit-aware patch validation outside the competition artifacts; the PoV-as-oracle literature begins and largely ends with AIxCC artifacts [46]. (ii) *Regression and redeployment*: B4 exists in our corpus solely as foundation-documented ecosystem transition — OpenSSF SIG hosting, \$200K-per-team integration funding, Team Atlanta’s \$2M prize recycling into continuous hunting [33], [34], [36] — never as an academic evaluation target with regression metrics. (iii) *Attacker adaptation*: no

included work studies adversaries reacting to deployed auto-patchers; the offensive side of B3 is absent. (iv) *Interaction with discovery*: patching and discovery are evaluated as separate competencies everywhere except inside CRS pipelines; whether autonomous discovery changes patch demand or maintainer workload at ecosystem scale is unmeasured (DARPA has not yet released the telemetry that would permit it [34]). These gaps seed the roadmap experiments of Sec. 11.

8. The Co-Evolution Loop: Graded Evidence

Our second insight concerns coupling between offense and defense. We grade every candidate edge on a three-level scale — *documented response* (an explicit, sourced statement or traceable institutional flow), *plausible coupling* (mechanism-reasonable temporal adjacency), and *temporal association* (chronology alone) — and claim only what the grades support: multiple documented edges exist; field-wide causal co-evolution does not follow from them.

8.1. Documented Response Edges

(D1) Competition → industrial research program. Project Zero’s Big Sleep post opens its target rationale by citing the challenge directly: “Earlier this year at the DARPA AIXCC event, Team Atlanta discovered a null-pointer dereference in SQLite, which inspired us to use it for our testing” [1]. This is the corpus’s cleanest causal edge: named source, stated mechanism, dated. **(D2) Maintainer-side variant response.** After Atlantis reported the FTS5 trigram off-by-one, the SQLite maintainer patched not only the reported function but sibling tokenizer code paths sharing the pattern, then revised his own patch in a follow-up commit after noting redundancy [35] — human variant analysis triggered by an AI finding. **(D3) Institutional/funding flows.** The loop acquired governance: DARPA and ARPA-H committed \$200K per team to integrate CRSs into critical software [34]; Team Atlanta recycled half its \$4M prize (\$2M) into continuous autonomous hunting at Georgia Tech’s SSLab with OpenAI contributing API credits [36]; OpenSSF chartered a Cyber Reasoning Systems SIG and hosts OSS-CRS and FuzzingBrain, whose post-competition operation is reported at 62 vulnerabilities across 26 projects (43 maintainer-confirmed, 36 patched upstream) for FuzzingBrain and 25 vulnerabilities across 16 projects for OSS-CRS [33]. These are foundation-relayed, team-reported figures; the funding and governance mechanisms provide evidence that competition-derived capabilities persisted beyond the scored event. **(D4) Threat-seeded defensive disclosure.** Google’s CVE-2025-6965 disclosure couples its Threat Intelligence unit to Big Sleep: the flaw was “known only to threat actors,” and the company asserts an imminent exploit was cut off — “we believe this is the first time an AI agent has been used to directly foil efforts to exploit a vulnerability in the wild” [47], [48]. The threat-intelligence premise is unverifiable and the foiling

counterfactual; we cite it as documented defensive process, not verified attack prevention.

8.2. Plausible Coupling

(P1) Attacker-side adoption of agentic operation within roughly twelve months of defender agentic publicity: Anthropic reports (as a single-source threat-intelligence assessment) that GTG-1002 used Claude Code for 80–90% of an espionage campaign’s execution — “the AI autonomously discovered vulnerabilities in targets selected by human operators” [5], [49]. Directionality is unverifiable without adversary telemetry; shared capability tide remains a live alternative.

8.3. Temporal Association Only

Benchmark releases trailing industrial demos, press arms-race narratives, and leaderboard movements populate this tier; per our claim ledger they carry no inferential weight here.

8.4. Falsification Test and Record Reliability

The shared-cause objection — frontier-model releases independently advancing both sides — survives globally: it is defeated only for edges D1–D2, where stated mechanisms precede any model-release timing argument. We therefore do not generalize from the AIXCC–Google hub to the field. Record reliability itself becomes evidence, once accounting scopes are labeled precisely: \$29.5M denotes the cumulative program prize commitment announced in 2024, \$8.5M denotes the final competition pool per DARPA’s scoring guide [37], and OpenSSF reports \$30.5M ultimately distributed across both rounds [33] — distinct boundaries, not competing totals for one quantity. Separately, DARPA’s own closeout corrects its earlier synthetic-vulnerability count from 70 to 63 [3]: an administrative record correction showing that even authoritative primary sources require version- and date-aware verification.

8.5. Institutional Response

Between one-off response and field-level causation sits a grade our corpus documents richly:
institutionalization. Three funding and governance flows carry competition-derived capability
past the scoreboard. DARPA and ARPA-H committed \$200K per finalist team to integrate CRSs
into critical software [34]. Team Atlanta recycled half of its \$4M prize
into continuous autonomous hunting of OSS-Fuzz projects at Georgia Tech’s SSLab, with OpenAI contributing API credits [36]. And OpenSSF chartered a Cyber Reasoning Systems
Special Interest Group that now hosts OSS-CRS and FuzzingBrain, relaying post-competition

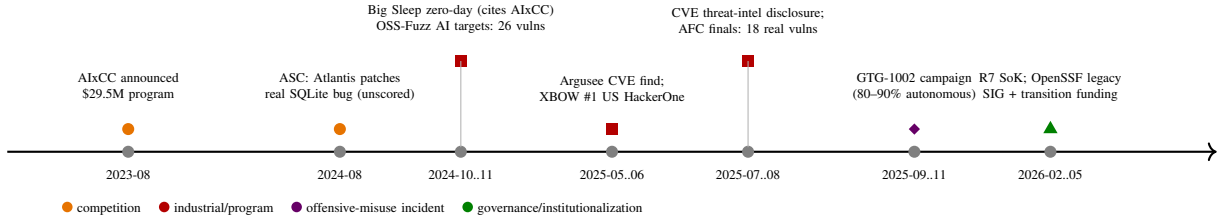


Figure 3: Capability-and-coupling timeline for the graded edges of Sec. VIII. Colors separate evidence classes; every event carries a primary source in Appendix C. The sequence visualizes I2’s documented edges — most notably the AIxCC→Big Sleep inspiration link — without implying field-wide causal closure.

yields — sixty-two vulnerabilities across twenty-six projects for FuzzingBrain

(forty-three maintainer-confirmed, thirty-six patched upstream), twenty-five vulnerabilities

across sixteen projects for OSS-CRS, and twelve kernel-related plus ten user-space zero-day

findings for 42-b3yond-6ug [33]. These counts remain team-relayed; the

institutions, budgets, and standing governance are independently observable. Google’s parallel

commitment — deploying Big Sleep against widely used open-source projects [47]

— extends the same pattern from competition lineage to industrial program.

On the speculative side of the ledger sits attacker-side adoption. The strongest candidate is

GTG-1002: a framework wrapping Claude Code with MCP tool access, jailbroken through task

decomposition and fabricated defensive-employment framing, executing reconnaissance,

exploit drafting, credential harvesting, and exfiltration with humans selecting targets and

retaining strategic authorization at critical escalation points [5], [49]. Its

grade stays *plausible coupling*: no adversary-side telemetry exists to date whether

defender publications caused, accelerated, or merely coincided with the technique transfer.

A concrete falsifier structures future work: if adoption-lag measurement finds agentic

offensive techniques appearing no earlier after major defender disclosures than baseline

diffusion rates would predict, response-coupling loses its offensive half and I2 narrows to

defender-internal loops.

8.6. Summary

The loop is real where it is documented: a competition generated tooling whose inspiration seeded an industrial program (D1), whose findings triggered maintainer responses (D2), whose participants now operate continuously under foundation governance (D3), and whose defender capability claims now interleave with threat-intelligence-driven disclosure (D4). What the record cannot yet support is the stronger

vernacular claim of an ongoing arms race; that requires adversary-side measurement this corpus lacks (Sec. 11).

9. Evaluation Validity

Our third insight treats validity as a measurable property of evaluations, not a limitations paragraph. We organize the evidence as a progression — information leakage, then dataset-level invalidity, then oracle manipulation — because each tier changes what a reported number can mean.

9.1. V2/V3: Information Leakage, Measured by Intervention

The cleanest datum in our corpus is controlled. Fang et al. measure GPT-4-based agents on 15 real one-day vulnerabilities under two information conditions: with the CVE description supplied, the agent autonomously exploited 87%; without it, success fell to 7%, while every baseline (GPT-3.5, eight open-source models, ZAP, Metasploit) scored zero in both conditions [6]. This more-than-an-order-of-magnitude difference is produced by a single supplied-knowledge variable — the CVE text functions as a partial answer key — so any capability claim inherits its information condition. We therefore report such results only with their conditions attached, and the same discipline applies to temporal leakage: every vulnerability in that benchmark was publicly disclosed before model training cutoffs.

9.2. V1/V5: Dataset-Level Invalidity, Measured by Reconstruction

PrimeVul demonstrates that the detection literature’s benchmark layer was structurally unsound: label noise, near-duplicate pairs across splits, and pre-split public exposure inflated reported performance to the point where paired, chronologically split evaluation collapses strong-model accuracy toward chance [11]. Risse and Böhme reach a compatible verdict from measurement theory, arguing that popular vulnerability datasets score models on the wrong exam [12]. The corpus also contains a live saturation illustration: Argusee reports 100% on CyberSecEval 2 buffer-overflow cases while its own documentation requires

human-supplied entry points [28] — ceiling scores coexisting with bounded autonomy. Agentic CTF suites inherit a further V1 exposure: their tasks derive from publicly written-up competitions, and no included work quantifies the inflation this induces; we return to it as roadmap experiment E2.

9.3. V4/V5: Oracle Manipulation and Environment Brittleness, Documented Operationally

AIxCC supplied operational evidence on outcome definitions. PoV re-execution establishes behavior only for the tested proof; teams documented mitigation-gaming patches — recompiling with MTE/PAC or wrapping entry points in broad *catch* blocks — that pass PoVs without semantic repair, prompting optional functional tests as secondary oracles and an accuracy modifier that made patch correctness the binding score term [36]. Environment leakage is exemplified by the winning CRS nearly disabling itself when organizers prefixed challenge directories with “ossfuzz,” breaking a “fuzz”-substring heuristic — load-bearing benchmark-specific logic invisible until the environment shifted [36]. On the production side, leaderboard-style counting conflates heterogeneous outcomes: of XBOW’s ~1,060 submissions, roughly 39% resolved into duplicate or informative categories rather than new/resolved findings — categories that reflect discovery overlap and program policy as much as agent error, making rank non-decomposable without category-aware accounting [2].

9.4. Administrative Record Correction: A Separate Category

DARPA’s closeout corrects its own synthetic-vulnerability count from 70 to 63 with an editor’s note [3], and prize accounting requires three scoped figures rather than one (\$29.5M cumulative commitment, \$8.5M final pool, \$30.5M distributed [33], [37]). These are administrative record corrections, not measurement-validity failures, but they carry a methodological moral we adopt as policy: even authoritative primary sources require version- and date-aware verification, and press intermediaries inherit whatever instability the primary contains.

9.5. A Contamination-Control Protocol

As a concrete deliverable, we specify the minimal protocol any agentic capability evaluation should publish: (i) *temporal firewall* — freeze the task set at date T , evaluate only models whose training cutoffs precede $T - \Delta$ for disclosed Δ , and release the task set only after evaluation; (ii) *provenance screening* — machine-check every task against public write-up corpora and NVD text up to T , reporting overlap rates rather than asserting cleanliness; (iii) *information-condition disclosure* — state exactly which artifacts (descriptions, patches, stack traces, credentials) are

supplied to the agent, Fang-style; (iv) *category-aware outcomes* — report validated/exploitable/duplicate/informative as separate counts with adjudication rules fixed before running; and (v) *adjudication audit* — a second scorer re-labels a random 10% sample with agreement statistics. No included evaluation satisfies all five today; E3 (Sec. 11) operationalizes the gap.

9.6. The Measured Core, Quantified

Three numbers anchor the progression above; each survives re-examination better than the headline claims it corrects. Fang et al.’s intervention separates information supply from capability: Pass@5 of 86.7% and an overall exploitation rate of 40% for GPT-4 under description-supplied conditions collapse to 7% overall without descriptions, while GPT-3.5, eight open-source models, ZAP, and Metasploit score zero throughout [6] — the condition, not the model, carries the result. PrimeVul’s reconstruction moves a state-of-the-art 7B detector from 68.26% F1 on BigVul to 3.09% on its paired, chronologically split evaluation, with GPT-3.5- and GPT-4-class prompting “akin to random guessing” under the most stringent settings [11] — dataset construction, not model capacity, had been the binding constraint on prior comparisons. And XBOW’s funnel makes outcome-definition noise concrete at production scale: of ~1,060 submissions, 208 duplicates and 209 informative reports — roughly 39% — resolve into neither new nor fixed findings, so leaderboard position aggregates discovery quality with program-policy artifacts unless category-aware accounting is applied [2]. Each datum predates or operates independently of the others; their convergence on the same methodological conclusion is why we treat validity as measured rather than alleged.

9.7. What Survives

Validity constraints do not render the corpus uninterpretable. Condition-controlled results survive (the 7% no-description condition), third-party adjudication survives (Ada Logics’ reproduction of 27 AIxCC-derived issues [33]), and platform triage provides weak external checks on vendor claims. What does not survive is unqualified headline numbers — which are precisely the ones that travel furthest.

10. Threats to Validity

Publication and visibility bias. Our industrial stream consists of organizations that *publish*: Google, Anthropic,

XBOW, DARKNAVY, OpenSSF. Systems that operate silently — commercial pentesting vendors, state programs, internal red teams — are structurally invisible, so our A3-empty finding is a statement about documented evidence, not about private capability. Proprietary disclosure is incomplete even where marketing is not.

Survivor and selection bias. Vendors report successes; failed autonomous deployments rarely generate blog posts. XBOW’s funnel (208 duplicates, 209 informative) is published only because its denominator was large enough to survive; smaller negative results have no venue.

Source-quality asymmetry. Academic records carry methods sections; vendor records carry dashboard screenshots. We mitigate with quote-level extraction, cross-source corroboration where possible (AIXCC cost tables agree across participant and systematization primaries), and explicit CLAIMED labels — but asymmetry remains, and one vendor-reported incident (GTG-1002) materially shapes the autonomy frontier of our corpus.

Rapid model churn. Capabilities tracked here span Gemini 1.5 Pro through mid-2026 models; any conclusion phrased about “LLMs” generally is hostage to a moving baseline. We therefore phrase findings about *records and their conditions*, not about model classes.

Incomplete benchmark access. Publisher APIs (IEEE, ACM) were unavailable during corpus construction; coverage leans on arXiv, DBLP, official indexes, and citation chaining. The field’s center of gravity publishes on those channels, but systematic gaps cannot be excluded, and screening operated on first-page result sets per query (documented in Sec. 3).

Subjective classification. Autonomy assignment under the decision-authority test requires judgment at boundaries. We bound it with two rules — classify by documented behavior and by the weakest missing condition — and we publish every record’s behavioral evidence in the artifact CSV; inter-rater reliability was not measured, which we flag as a limitation.

Temporal ambiguity. Several artifacts describe systems whose capability changed between event and publication (XBOW’s leaderboard state; Big Sleep’s model version). We date claims to their sources’ datelines and mark snapshot statistics as such.

Undisclosed human assistance. Platform policies mandate review only where platforms exist; nothing prevents undisclosed human effort behind vendor claims of full automation. Our classifications therefore track *documented* autonomy floors, not ceilings.

Single-corpus induction. All quantitative distributions (14 autonomy-bearing records; per-cell counts) derive from one team’s inclusion decisions under one protocol; the saturation test bounds but does not eliminate corpus-dependence. Where a claim would not survive a different reasonable inclusion rule, we say so in place.

Benchmark contamination. Recent work demonstrates that LLM training data often includes benchmark tasks and their solutions [11], [30]. Our validity taxonomy (V1–V5) names this threat explicitly, and we adopt the five-

part contamination-control protocol (Sec. 9) as a concrete mitigation recommendation. However, we cannot guarantee that any included benchmark result is contamination-free; this limits the strength of capability claims derived from benchmark pass rates.

Absence of adversarial evaluation. Our corpus contains no red-team evaluation of the systems we classify. We cannot assess whether the A2 systems we describe would survive adversarial probing designed to find failure modes, edge cases, or exploitation paths that standard benchmarks do not cover. This is a structural limitation of working from published artifacts rather than conducting independent evaluation.

Ethics.. This study is documentary: we analyzed published artifacts and did not execute exploits, contact vendors, or handle vulnerability data beyond what primary sources already disclose. Where our corpus touches responsibly disclosed vulnerabilities, identifiers are handled per venue policy, and no new attack capability is introduced beyond quoted primary material.

11. Open Problems: An Experimental Roadmap

Every gap below is stated with a falsifier: the observation that would disconfirm the expected result. We prioritize experiments by how efficiently they resolve load-bearing uncertainties identified in Secs. 5–9.

E1 — Isolating autonomy from model scale (I1’s unresolved half). Gap: no included work ablates autonomy-level contribution from base-model improvement. Setup: fix a task suite and environment; evaluate the same base model across A1-style fixed pipelines and A2-style dynamic agents, repeated across successive model releases. Controls: identical tools, prompts budgets, and scoring; release-pinned checkpoints. Metric: marginal success attributable to decision authority vs. model generation. Falsifier: if fixed pipelines close the gap to agents as models improve, “autonomy” is re-branding of scale, weakening I1’s substantive reading.

E2 — Quantifying write-up exposure in agentic CTF suites (V1). Gap: public-solution contamination is widely hypothesized but insufficiently quantified for multi-step agents. Setup: time-split evaluation on challenges disclosed after each model’s cutoff, against a matched pre-cutoff set. Controls: difficulty matching via human solve rates; retrieval audits. Metric: within-model success delta across the disclosure boundary. Falsifier: near-zero delta implies CTF benchmarks measure general problem solving, not leaked knowledge — and current headline numbers stand.

E3 — Contamination-controlled agentic benchmark (§9 protocol instantiated). Gap: no included evaluation satisfies all five protocol requirements. Setup: build a benchmark under the five-part protocol (temporal fire-wall, provenance screening, information-condition disclosure, category-aware outcomes, adjudication audit). Con-

trols: release pipeline auditable from task creation. Metric: fraction of prior-suite rankings preserved. Falsifier: high rank correlation with contaminated predecessors would show contamination is not distorting conclusions, bounding I3.

E4 — Exploit-aware patch validation. Gap: PoV-passing patches may be semantic non-fixes (mitigation gaming). Setup: differential testing of AIxCC-derived patches under alternate exploit inputs, sanitizer diversity, and semantic-equivalence checks. Controls: human-expert ground truth on a sample. Metric: semantic-repair rate vs. PoV-pass rate. Falsifier: if the rates coincide, PoV oracles suffice and the gaming taxonomy is a curiosity.

E5 — The patch/retest loop at ecosystem scale (B4). Gap: B4 exists only as foundation-documented transition; regression impact unmeasured. Setup: instrument CRS-vs-maintainer workflows on live OSS projects (the OpenSSF-hosted tools provide the fleet); track maintainer latency, acceptance, regressions. Controls: pre-registration; maintainers blinded to tool origin. Metric: time-to-fix and regression rate deltas. Falsifier: null effect would show competition-derived tooling did not alter ecosystem patch dynamics, trimming D3’s interpretation.

E6 — Measuring adversary-side response. Gap: co-evolution evidence is defender-side; attacker responses are anecdotal. Setup: longitudinal measurement of attacker tooling/behavioral markers in incident telemetry shared under defense-pact agreements. Controls: differential exposure to defender publications. Metric: adoption lag of agentic techniques post-disclosure. Falsifier: absence of lag structure supports shared-cause over response-coupling, capping I2 at its current grade.

E7 — Decision-impact audit (upgrading I3). Gap: “binding constraint” lacks a documented case of a decision changing on validity correction. Setup: retrospective audit of rankings, funding calls, and procurement that consumed pre-PrimeVul-era results; interview plus document tracing. Metric: count of decisions reversed or re-scored after validity correction. Falsifier: zero documented cases would cap I3 at “important constraint” and justify removing “binding.”

12. Conclusion

This systematization asked whether LLM-driven vulnerability work between January 2024 and June 2026 is best understood through model capability or through something else. Our answer, built on a 36-record multi-stream corpus of which 14 records bear autonomy classifications, is that the field’s evaluations, incentives, and discourse now measure progress in *agentic autonomy* — while whether underlying capability moved for reasons other than scale remains, honestly, an open question.

On I1 (established in framing; unresolved in substance). The autonomy gradient A0–A3 organizes the corpus cleanly: capability probes (A0) persist only as benchmark subjects; pipeline automation (A1) runs at ecosystem scale with humans at review boundaries; task agents

(A2) appear across seeded production research, entry-point-assisted auditing, platform-operated pentesting, and competition CRSs integrating discovery, proof, and patching. What the record does not contain — under our four-verb, environment-E2 operational definition — is a single A3 autonomous production hunter. We state the finding only as: no included evidence met our operational definition of A3. The nearest candidate is an offensive incident, vendor-reported with hallucination caveats and human-selected targets.

On I2 (documented edges; causality bounded). Offense and defense are coupled where the record says so: Project Zero named AIxCC’s Team Atlanta as the inspiration for its SQLite variant-analysis campaign; a maintainer extended an AI-found bug’s fix to sibling code paths; \$29.5M in cumulative program prizes funded \$200K/team transition integration and university research, while \$30.5M in competition distributions supported foundation governance; and threat intelligence seeded a pre-exploitation disclosure. These documented edges do not license the arms-race vernacular: shared-cause explanations survive globally, and attacker-side response remains unmeasured.

On I3 (observed; binding claim pending E7). Validity failures here are not rhetorical. Information conditions change reported exploitation rates by more than an order of magnitude; dataset reconstruction collapses detection SOTA; oracle gaming has a named taxonomy; environment-specific heuristics can materially affect system outcomes; and leaderboard counting conflates outcome categories. Because these measures allocate prizes and rank vendors directly — and inform funding decisions through transition programs — evaluation validity *appears to be* a binding methodological constraint on what this field can claim to know, though the decision-impact audit (E7) has not yet been conducted to confirm this assessment.

What we ask of the next iteration of this literature. Publish information conditions with every capability number; adopt category-aware outcome accounting; evaluate patching against exploit-aware oracles; measure B4 rather than narrate it; and treat autonomy claims by the weakest missing condition, not the strongest available adjective. The tools described in this paper will be judged — fairly or not — by whether they make those practices easier.

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Appendix A.

Included Records: Classification, Quality, and Provenance

Complete per-record audit artifact: classification (axes per Sec. IV), quality rubric (reproducibility/baseline-rigor/artifact availability, 0–3), evidence class, and access provenance. Behavioral evidence per record ships in `evidence/autonomy-loop-assignments.csv`.

Appendix B.

Search Log Summary

Stream A queries (arXiv UI, window 2024-01-01..2026-06-30, submitted-date filter): Q1 “large language model” AND “vulnerability detection” (253); Q2 “language model” AND “vulnerability discovery” (42); Q3 “language model” AND “penetration testing” (67); Q4 “language model” AND “CTF” (36); Q5 “LLM agent” AND “exploit” (192); Q6 “large language model” AND “program repair” (202); Q7 “cyber reasoning system” (9); Q8 “AI agent” AND “vulnerability” (172); subtotal 973 raw hits; deduplicated first-page screening union 316 distinct IDs. DBLP API: D1 “large language model” vulnerability (174); D2 “large language model” “penetration testing” (11); D3 LLM CTF benchmark (5); D4 vulnerability repair LLM (24); subtotal 214. Streams B/C: targeted retrieval with per-source verification entries in the artifact manifest. Executed 2026-08-26; full strings and failure notes in repository file `search-log.md`.

Appendix C.

AIXCC and Industrial Timeline

TABLE 5: Dated event chain (primary sources where available; secondary synthesis explicitly identified; grades per Sec. 8).

Date	Event (source)
2023-08	AIXCC announced; ~\$29.5M cumulative program commitment [3]
2024-08	ASC: 42 teams → 7 finalists (\$2M each); Atlantis autonomously patches real SQLite FTS5 bug (unscored); official fix e9b919d5 follows [35]
2024-11	Big Sleep reports SQLite stack-buffer-underflow zero-day; cites AIXCC inspiration (edge D1); AFL 150 CPU-h fails to rediscover; no CVE assigned [1]
2024-11	OSS-Fuzz reports 26 vulns from AI-generated fuzz targets incl. CVE-2024-9143 (~20 years latent) [27]
2025-05	DARKNAVY Argusee finds CVE-2025-37891 (Linux USB); root LPE by human team [28]
2025-06	XBOW #1 US HackerOne; ~1060 submissions; mandated human review disclosed [2]
2025-07	CVE-2025-6965 threat-intel-seeded disclosure; “first time an AI agent has been used to directly foil” claim [47], [48]
2025-08	AFC finals: 48 challenges/23 repos; 143 h fully autonomous operation; 18 real vulns found; Atlanta \$4M / ToB \$3M / Theori \$1.5M; \$200K/team transition funding [3], [34], [36], [37]
2025-09..11	Anthropic reports GTG-1002 campaign, 80–90% autonomous execution [5], [49]
2026-02..05	R7 SoK camera-ready (USENIX Sec’26); OpenSSF legacy retrospective: CRS SIG, FuzzingBrain 62v/26proj, OSS-CRS 25v/16proj [4], [33]

TABLE 4: All 36 included records. A =autonomy level; B =loop-position set; R/B/Ar = rubric scores; Acc = access status.

Record	Year	Unit	A	B -set / role	Evidence class	R/B/Ar	Acc
<i>In-window autonomy-bearing records (14)</i>							
agentless2024	2024	system(experiment-benchmark)	A1	B3	experiment	3/3/3	PDF
argusee2025	2025	system	A2	B1	production discovery	0/2/0	HTML
artemis2025	2025	experiment	A2	B1+B2	production-network experiment	2/3/2	PDF
autosolverever-2024	2024	system(experiment-benchmark)	A2	B3	experiment	3/3/3	PDF
bigsleep2024	2024	system	A2	B1	production discovery	0/2/0	HTML
fang2024one-day	2024	experiment	A2	B2	controlled experiment	2/3/3	PDF
gtg1002-2025	2025	incident-documentation	uncertain	B1+B2	incident report	0/0/0	HTML
ossfuzz-aifxing-2024	2024	experiment	A2	B3	experiment	2/2/2	PDF
ossfuzz-levelingup-2024	2024	system(pipeline)	A1	B1	industrial-system-report	1/1/2	HTML
sweagent2024	2024	system(experiment-benchmark)	A2	B3	benchmark/experiment	3/3/3	PDF
teamatlanta-aic2025	2025	system	A2	B1+B2+B3	competition-participant	1/2/2	HTML
teamatlanta-sqlite2024	2024	system	A2	B1+B2+B3	competition-participant	1/2/2	HTML
tob-butercup2025	2025	system	A2	B1+B2+B3	competition-participant	1/3/2	HTML
xbow2025	2025	system	A2	B1+B2	production stats	0/1/0	HTML
<i>Background anchors (excluded from primary counts)</i>							
csev12023bg	2023	benchmark	A0	B1	benchmark	2/1/2	PDF
intercode2023bg	2023	benchmark	A1	B2	benchmark	3/2/3	PDF
<i>Loop-transition documentation (B4)</i>							
bloggoogle-summer2025	2025	program-record	n/a	B4 — documents B1 discovery -> announced B4 redeployment of Big Sleep to OSS projects	program-announcement	—/—/—	HTML
openssfretro2026	2026	program-record	n/a	B4 — documents post-competition B3->B4 institutionalization (SIG hosting, transition funding, ecosystem stats)	foundation closeout	1/1/2	HTML
<i>Off-axis / context records</i>							
csev22024	2024	benchmark	n/a	n/a — probed B1,B2	benchmark	2/2/2	PDF
csev32024	2024	benchmark	n/a	n/a — probed B1,B2	benchmark	2/2/2	PDF
cybench2024	2024	benchmark	n/a	n/a — probed B2	benchmark	3/3/3	PDF
mcpeco2025	2025	measurement-study	n/a	n/a	measurement	2/2/3	PDF
mcpgov2025	2025	documentary-source	n/a	n/a	positional/measurement	1/1/2	PDF
mcpoison2026	2026	documentary-source	n/a	n/a	analysis/experiment	2/2/2	PDF
mcpspec2026	2026	documentary-source	n/a	n/a	analysis	2/2/2	PDF
nvdcve20256965	2025	registry	n/a	n/a	registry	3/3/3	HTML
nyuctfbench2024	2024	benchmark	n/a	n/a — probed B2	benchmark	3/2/3	PDF
primevul2024	2024	dataset	n/a	n/a — B1 detection (dataset standard)	dataset/validity study	3/3/3	PDF
r1slr2024	2024	survey	n/a	n/a	survey	1/1/0	PDF
r2slr2024	2024	survey	n/a	n/a	survey	1/1/0	PDF
r3survey2026	2026	survey	n/a	n/a	survey	1/2/0	PDF
r4csur2025	2025	survey	n/a	n/a	survey	0/0/0	payw.
r5usenix26agentic	2026	survey	n/a	n/a	survey	0/0/0	PDF
rfotosem2026	2026	survey	n/a	n/a	survey	0/0/3	payw.
r7soikaixcc2026	2026	survey(systematization)	n/a	n/a	competition	2/3/2	PDF
rissebohme-wrongexam2024	2024		n/a	n/a	validity-critique	—/—/—	PDF